LEXAI PROJECT: 4-MEMBER ROLE-BASED CASE STUDY PORTFOLIO

Natural Language Processing Project — AI Legal Assistant (LexAI)

Department: Computer Science and Engineering | **Academic Year:** 2025–2026

■ Portfolio Directory & Individual Case Studies

Team Member	Roll Number	Specialized Project Role	Primary Ownership Domain	Case Study Document Link
V. Brunda	24B11CS485	**Lead NLP Pipeline & Machine Learning Engineer**	Preprocessing, Tokenization, Domain Classifier (100% Acc), Portable Weights Export	View Case Study 1
G. Krishna	24B11CS127	**Legal Risk Classification & Information Extraction Specialist**	Clause Risk Stratification (92.81% Acc), 53 Legal Categories, Centroid Summarizer	View Case Study 2
G. Jahnavi	24B11CS512	**Backend Architecture & RAG Pipeline Engineer**	Node.js Express API, Multi-Format Ingestion (PDF/DOCX), RAG Vector Store, Security	View Case Study 3
K. Mamatha	24B11CS229	**Full-Stack Frontend & Legal UX/UI Engineer**	React 18 SPA, Dynamic 0–100 Risk Gauge, Clause Accordion Audit, Drafting Wizard	View Case Study 4

■■ Complete System Architecture & Role Interaction

```
graph TD
    subgraph Client Tier [Role 4: K. Mamatha - Frontend & UX]
        UI[React 18 + Vite SPA]
        Gauge[Dynamic SVG Risk Gauge 0-100]
        Inspector[Clause-by-Clause Audit Inspector]
        Wizard[Legal Document Drafting Wizard]
    end

    subgraph Server & RAG Tier [Role 3: G. Jahnavi - Backend & RAG Engine]
        API[Node.js / Express REST API]
        Stream[Zero-Disk Memory Stream Ingestion]
        RAG[Statutory RAG Retriever & Vector Store]
        Gate[Strict Anti-Hallucination Confidence Gating]
    end

    subgraph Document Intelligence Tier [Role 1: V. Brunda - NLP & ML Pipeline]
        Preproc[NFKD Normalization & Deontic Tokenizer]
        DocModel[Document Domain Classifier: Sublinear TF-IDF + Logistic Reg]
        Weights[Portable JSON NLP Weights Exporter]
    end

    subgraph Risk Auditing Tier [Role 2: G. Krishna - Risk & Entity Specialist]
        Segment[Boundary Lookbehind Clause Segmentation]
        RiskModel[Clause Risk Classifier: High / Medium / Low]
        TypeModel[53-Class Fine-Grained Clause Taxonomy]
        Summ[Centroid Salience Extractive Summarizer]
    end

    UI -->|Multipart Upload| API
    API --> Stream
    Stream --> Preproc
    Preproc --> DocModel
    DocModel --> Segment
    Segment --> RiskModel
    Segment --> TypeModel
    Segment --> Summ
    RiskModel & TypeModel & Summ --> API
    API --> RAG
    RAG --> Gate
    Gate --> API
    API -->|Structured Audit JSON Payload| UI
    UI --> Gauge & Inspector & Wizard
```

■ Consolidated Technical Benchmark Summary

Evaluation Dimension	Metric / Benchmark	Owner	Output & Validation Status
Document Classification	**100.00% Accuracy / 1.0000 F1**	V. Brunda	0 errors across 4 domains (House, Car, Bank, Gold)
Clause Risk Classification	**92.81% Test Acc / 90.53% CV F1**	G. Krishna	Evaluated over 5,281 labeled Indian legal clauses
Clause Taxonomy (53 Types)	**87.76% Acc / 87.62% Weighted F1**	G. Krishna	Multi-class LinearSVC with balanced weighting
Extractive Summarization	**0.00% Hallucination Rate**	G. Krishna	Deterministic TF-IDF centroid sentence selection
Native Inference Latency	**11.8 ms Execution Time**	G. Jahnavi & V. Brunda	Pure JS portable weights running in Node.js runtime
PDF / DOCX Ingestion SLA	**< 420 ms End-to-End Latency**	G. Jahnavi	In-memory stream buffer parsing (`pdf-parse`, `mammoth`)

Statutory Grounding	**100% Citation Verification**	G. Jahnavi	Grounded in BNS 2023, Transfer of Property Act, COPRA
Frontend Core Web Vitals	**Lighthouse 98/100, FCP 0.62s**	K. Mamatha	Zero layout shifts, WCAG 2.1 accessible dark/light UI
Drafting Wizard Efficiency	**3.2 min vs 2.5 hrs manual**	K. Mamatha	Formats Section 35 COPRA complaints & Tenancy Deeds